

- Q.27 List any five advantages of Emulsion Polymerization.
- Q.28 Define degree of Polymerization and write its importance.
- Q.29 Explain the structure and four uses of polypropylene.
- Q.30 Describe the preparation and four uses of polystyrene.
- Q.31 Describe the role of stabilizers in plastic.
- Q.32 Give the structure of polyvinyl chloride and write any four applications.
- Q.33 Explain injection molding.
- Q.34 State any five differences between low-density and high-density polyethylene.
- Q.35 Explain the structure and four properties of nylon6.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the technique of suspension polymerization and state its advantages and disadvantages (any five) over bulk polymerization.
- Q.37 With neat and labelled sketch, explain blow molding technique for polymer processing and name two articles produced by blow molding.
- Q.38 Write short notes on the following-
- Filler's
 - Application of additives

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4th Sem / Chem

Subject:- Polymer Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The number of repeating units in a polymer is known as -
- Monomer
 - Degree of polymerization
 - Chain
 - Molecules
- Q.2 Bakelite is manufacturing from-
- Phenol & formaldehyde
 - Ethylene glycol
 - Urea & Formaldehyde
 - None of these
- Q.3 Which of the following is thermosetting polymer?
- Terylene
 - Polystyrene
 - Bakelite
 - Neoprene
- Q.4 Which of the following is a polyamide polymer?
- Nylon6,6
 - Terylene
 - Teflon
 - Polythene

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- Q.5 The monomer of poly vinyl chloride (PVC) is-
- Chloroethane
 - Ethylene dichloride
 - Ethylene chloride
 - Chloroform
- Q.6 Plasticizers are added to synthetic plastics to-
- Impact flexibility
 - Improve workability during fabrication
 - Develop new improved properties not present in the original resin.
 - All (a), (b) and (c)
- Q.7 Which of the following is a copolymer?
- PVC
 - Bakelite
 - Polythene
 - Teflon
- Q.8 The process involved in converting rubber into a thin sheet or coating it on fabric is called-
- Extrusion
 - Molding
 - Calendaring
 - Vulcanisation
- Q.9 Chain-growth polymerization consists of how many major steps?
- 1
 - 2
 - 3
 - 4
- Q.10 Rayon is a _____ fiber.
- Cellulosic
 - polyamide
 - Polyester
 - None

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Give an examples of thermoplastic polymer.
- Q.12 Define monomer.
- Q.13 Write the formula of polyethylene.
- Q.14 Write one use of urea formaldehyde.
- Q.15 Give two examples of coloring material.
- Q.16 Write one free radical.
- Q.17 Write two articles produced by blow molding.
- Q.18 Explain initiators in polymerization.
- Q.19 Define Fiber.
- Q.20 Name one elastomer.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain step growth polymerization.
- Q.22 What is solution polymerization?
- Q.23 Explain briefly the five effects of molecular weight on properties of polymers.
- Q.24 Describe the concept of weight average molecular weight.
- Q.25 Discuss various method of chain termination.
- Q.26 Describe co-polymerization by giving an example.

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